ESERCIZIO n°1

$\int_{t\in\Sigma} \langle F, T\rangle ds \qquad F=(z,x,y^2) \qquad z=x^2+y^2 \qquad 1\le z\le 4$

$\begin{cases} x=\cos t \\ y=\sin t \\ z=x^2+y^2=1 \end{cases} \quad t\in[0,2\pi] \qquad T=(\cos t,\sin t,1) \qquad T'=(-\sin t,\cos t,0)$

INT. CURVILINEO

$\int_{t\in\Sigma}\langle(1,\cos t,\sin^2 t),(-\sin t,\cos t,0)\rangle ds = \int_{t\in\Sigma} -\sin t+\cos^2 t\, ds = \int_0^{2\pi} -\sin t+\cos^2 t\, dt = \int \frac{1+\cos(2t)}{2}\, dt = \frac{1}{2}\int 1+\cos(2x)\, dx = \pi$

INT. SUPERFICIALE

$$\text{ROT } F = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial x & \partial y & \partial z \\ z & x & y^2 \end{vmatrix}$$

$$\text{ROT } F = \left(\frac{\partial F_3}{\partial y}-\frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z}-\frac{\partial F_3}{\partial x}, \frac{\partial F_2}{\partial x}-\frac{\partial F_1}{\partial y}\right)$$

$\text{ROT } F = (2y-0, 1-0, 1-0) = (2y,1,1)$

$n_e = \frac{-f_x,-f_y,1}{\sqrt{1+f_x^2+f_y^2}} \longrightarrow \frac{-2x,-2y,1}{\sqrt{1+4x^2+4y^2}} \qquad d\sigma = \sqrt{1+4x^2+4y^2}\, dx dy$

$\iint_\Sigma \langle(2y,1,1)\cdot\left(\frac{-2x,-2y,1}{\sqrt{1+4x^2+4y^2}}\right)\rangle \sqrt{1+4x^2+4y^2}\, dx dy = \iint_\Sigma -4xy-2y+1\, dx dy$

$\begin{cases} x=r\cos t \\ y=r\sin t \\ z=1 \end{cases} \quad r\in[0,1] \quad t\in[0,2\pi] \qquad dx dy = dt\, dr\cdot|\det(\text{Jac})| \qquad dx dy = r\cdot dt\cdot dr$

$\int_0^1 dr\int_0^{2\pi}(-4\cos t\sin t-2\sin t+1)r\, dt = \int_0^1 dr\int_0^{2\pi} -4r\cos t\sin t-2r\sin t+r\, dt = \int_0^{2\pi}$

ESERCIZIO n°2

$\omega = (x^2+y)dx + x\,dy$

$\frac{\partial F_1}{\partial y}=1 \quad \frac{\partial F_2}{\partial x}=1 \longrightarrow \frac{\partial F_1}{\partial y}=\frac{\partial F_2}{\partial x}$ È CHIUSA

$\omega = \overbrace{(x^2+y)dx}^{F_1} + \overbrace{x\,dy}^{F_2}$

$\int x^2+y\, dx = \frac{x^3}{3}+xy+c(y)$

$\frac{\partial f}{\partial y} = x+c'(y)$

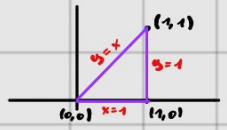
$x+c'(y)=x$

$c'(y)=0 \longrightarrow c(y)=0+c$ (INTEGRI)

$\boxed{\frac{x^3}{3}+xy+c}$ PRIMITIVA O POTENZIALE

ESERCIZIO n°3

$f(x,y)=x^2+8y^2+2x^2y$



$y=mx+q$

$\frac{\partial F_1}{\partial x}=2x+4xy \qquad \frac{\partial F_1}{\partial y}=16y+2x^2$

$\begin{cases} 2x+4xy=0 \longrightarrow 2x(1+2y)=0 \longrightarrow x=0 \\ 16y+2x^2=0 \longrightarrow 16y=-2x^2 \longrightarrow y=-\frac{x^2}{8}=0 \end{cases}$

$1+2y=0 \longrightarrow y=-\frac{1}{2}$

$16\left(-\frac{1}{2}\right)+2x^2=0 \longrightarrow x=\pm 2$

$\frac{8}{2}x^2 = \frac{8^4}{2^1}\Big|_{x=2}$

A(0;0) PUNTO CRITICO

B(2, ·) C(-2, ·)

- $F(x,y,\lambda)=f(x,y)+\lambda g(x,y)$

$g(x,y)=0 \longrightarrow y=x \longrightarrow y-x=0$

$F(x,y,\lambda)=x^2+8y^2+2x^2y+\lambda(y-x)$

$\nabla f=0 \longrightarrow \begin{cases} F_x'=\frac{\partial f}{\partial x} \\ F_y'=\frac{\partial f}{\partial y} \\ F_\lambda'=\frac{\partial f}{\partial \lambda} \end{cases} \Longrightarrow \begin{cases} 2x+4xy-\lambda=0 \\ 16y+2x^2+\lambda=0 \\ 2x+4x^2-\lambda=0 \end{cases}$